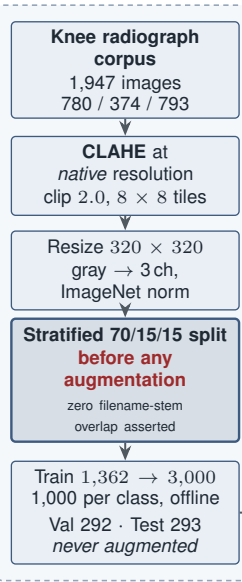
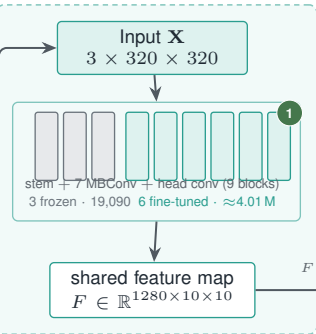


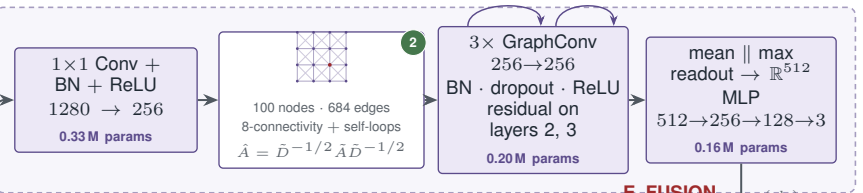
A DATA PREPARATION



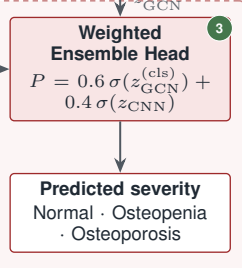
B EfficientNetB0 ENCODER



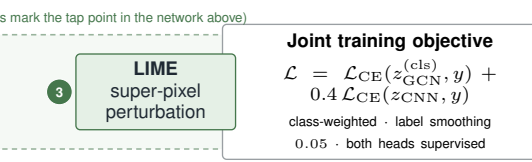
C PATCH-GRAPH CONSTRUCTION + RESIDUAL DUAL-POOLING GRAPH REASONING BRANCH



E FUSION



D AUXILIARY CONVOLUTIONAL BRANCH (graph-free path)



F THREE-TIER EXPLAINABILITY SUITE (numbers mark the tap point in the network above)



Joint training objective

$$\mathcal{L} = \mathcal{L}_{CE}(z_{GCN}^{(cls)}, y) + 0.4 \mathcal{L}_{CE}(z_{CNN}, y)$$

class-weighted · label smoothing
0.05 · both heads supervised